



HACK2IGNITE 2026



SPACELOOP

Turn any unused space into a temporary, verified, purpose-built micro-workspace.

TEAM NAME

LOGIC LOOP

PROBLEM STATEMENT ID

OPEN INNOVATION

INSTITUTION

GH RAISONI INTYERNATIONAL SKILL TECH UNIVERSITY

TEAM MEMBERS

MEMBER 01



INDRAYANI MAZUMDER

AI/ML / Research & Integration

MEMBER 02



KANISHK SINGH

Backend Developer and product designer

MEMBER 03



ZARA QUADRI

Frontend / UI-UX Developer

MEMBER 04

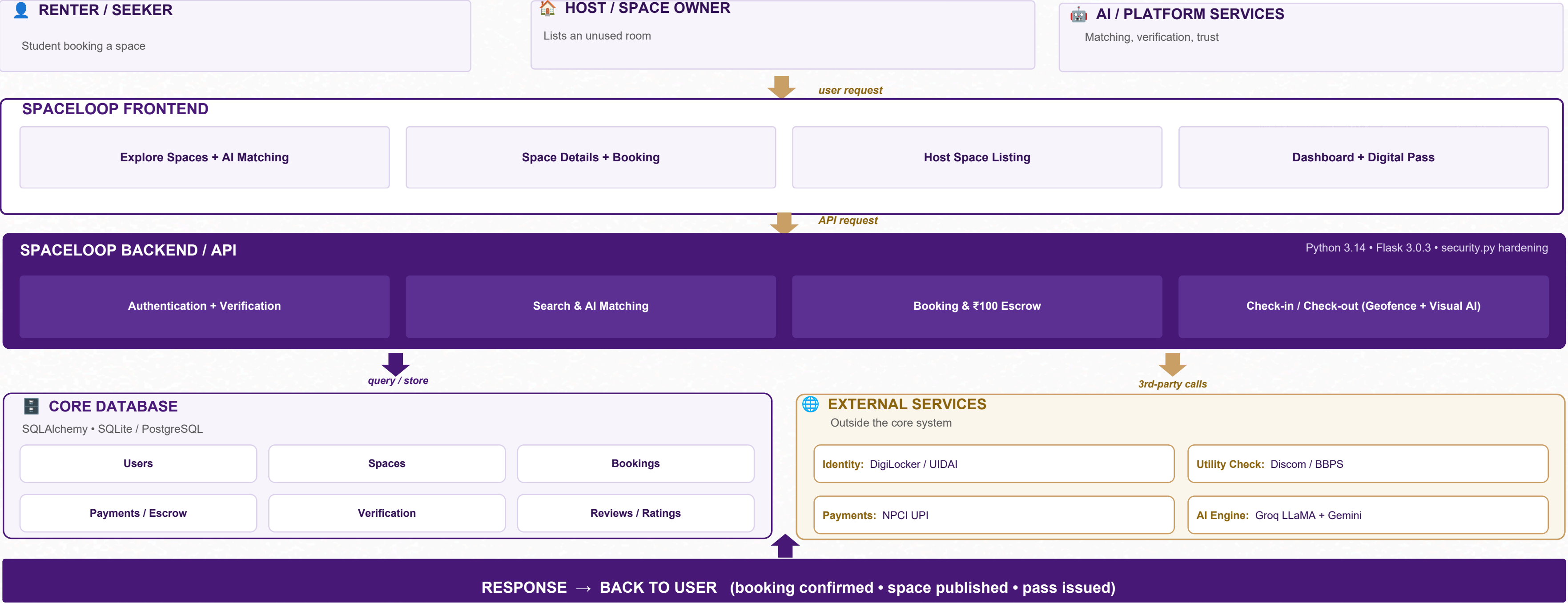


AARYA MAURYA

System Architect and Security

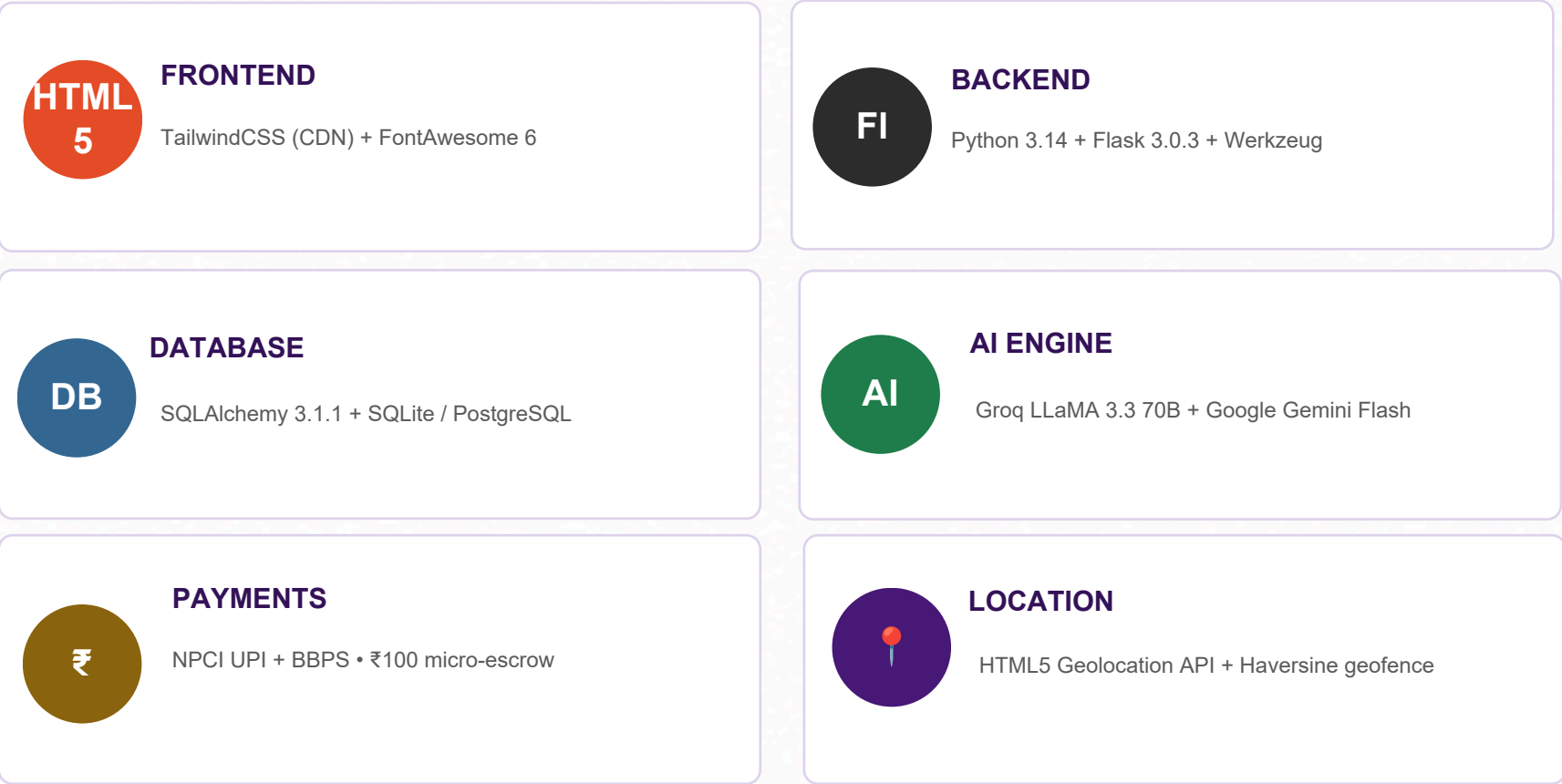
SYSTEM ARCHITECTURE

How SpaceLoop works, end to end

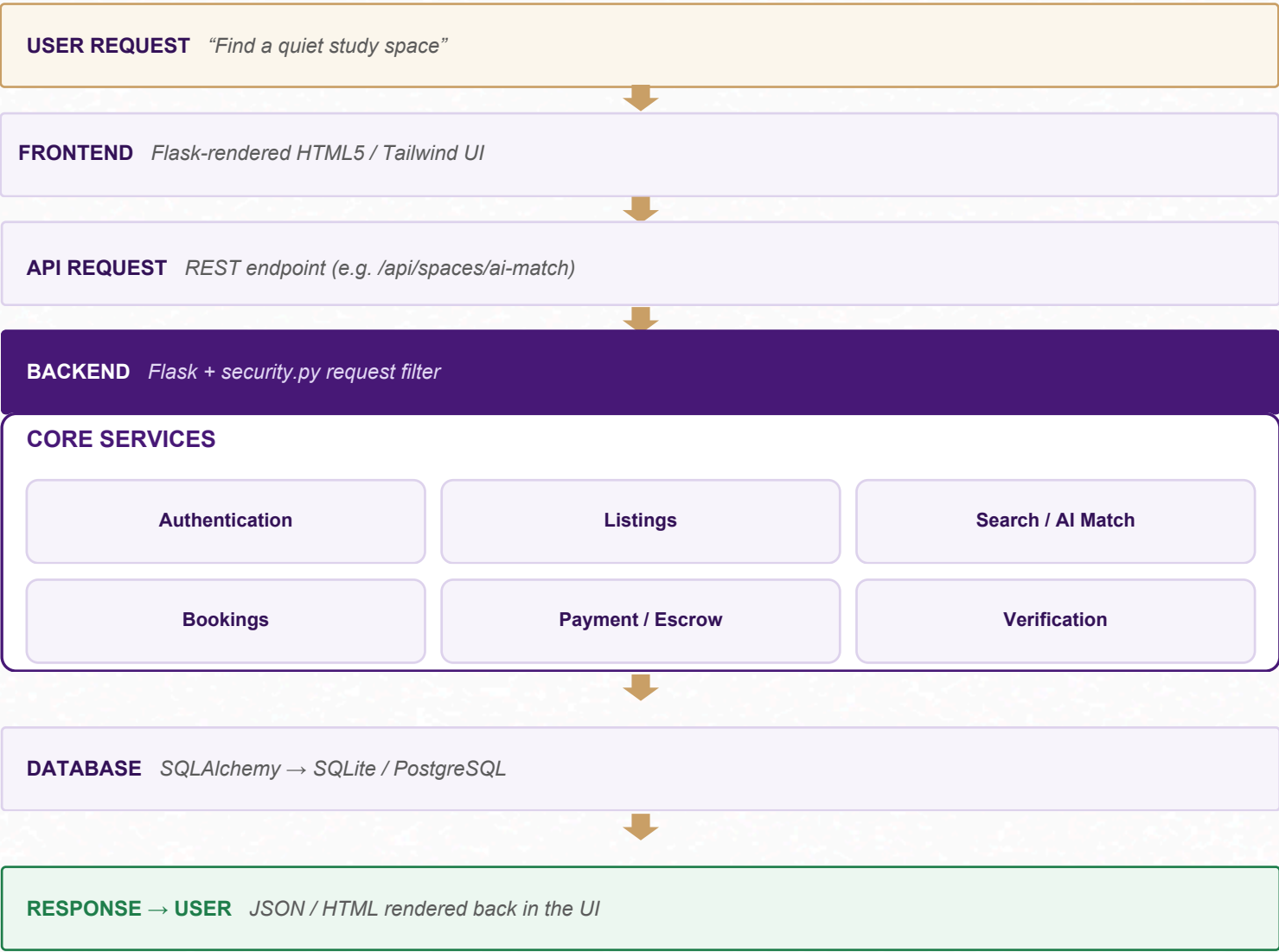


TECHNICAL APPROACH

TECHNOLOGY STACK



BACKEND FLOW — REQUEST LIFECYCLE



e.g. “Book this space” → Booking API → Payment / Escrow → Digital Pass → User

FEASIBILITY AND VIABILITY

MARKET FEASIBILITY

- Unused rooms exist across every dense urban / campus micro-market
- Students increasingly need flexible, hourly access, not full-day rent
- Traditional leasing demands high commitment hosts resist
- Hourly access lowers the commitment barrier for renters
- Hosts can monetize idle capacity with zero hardware spend

CHALLENGES

- Safety of the renter inside a stranger's space
- Trust between first-time hosts and renters
- Property damage during the session
- Physical access security without smart locks
- Payment disputes over deposits
- Local regulations around short-term tenancy

MITIGATION

- DigiLocker Aadhaar + institutional ID verification
- Objective Telemetry Index replaces fakeable star ratings
- Instant ₹100 UPI micro-escrow, programmatically released
- Discom CA + GPS geofence confirms the real, verified space
- QR pass + dynamic caretaker PIN controls physical access
- AI visual diff + Financial Clearance score resolves disputes

65% CHEAPER

than IoT smart-lock coworking — ₹40–₹75/hr on SpaceLoop vs ₹250–₹450/hr on hardware-based PropTech

IMPACT AND BENEFITS

HOSTS

- Monetize unused space
- Generate flexible passive income
- Reduce vacancy loss on idle rooms
- Improve utilization of existing property

RENTERS

- Affordable, short-term hourly access
- No long-term lease commitment
- Purpose-specific, verified spaces
- Flexible, on-demand booking

SPACELOOP

BETTER SPACE UTILIZATION

ECONOMIC IMPACT

New passive-income asset class from previously idle urban rooms

SOCIAL IMPACT

Affordable, safe, verified study & work access for students

URBAN EFFICIENCY

Existing built space reused instead of new commercial fit-outs

RESEARCH AND REFERENCES

INDUSTRY DATA

- Large stock of unused / underutilized residential & campus-adjacent rooms in Indian metros
- Rising cost of commercial coworking & smart-lock PropTech fit-outs
- Growing preference for flexible, pay-per-use workspace over fixed leases

MARKET TRENDS

- Gig economy & micro-entrepreneurship on the rise
- Flexible / shared workspace adoption growing in urban India
- Pop-up businesses and short-term rentals gaining traction
- [REFERENCE TO BE INSERTED — market sizing report]

PROJECT RESEARCH & SYSTEMS

- DPDP Act 2023 & UIDAI — Aadhaar tokenization compliance
- Indian Easements Act 1882, Sec. 52 — revocable license structure
- State Discom / BBPS: BESCOM, TPDDL/BSES, MSEDCL, UPPCL, Adani Electricity
- NPCI UPI penny-drop & escrow rails
- Groq LLaMA 3.3 70B & Google Gemini Flash (AI reasoning)

FUTURE INTEGRATIONS

- SpaceLoop can evolve into a production-grade **modular monolith with Next.js + TypeScript** on the **frontend** and **FastAPI + PostgreSQL** on the **backend**, using **Redis for caching**, sessions, rate-limiting and background queues, and OpenSearch for advanced location/AI-powered search. Add **S3/R2 for images and documents**, **Celery for background jobs**, **Docker + GitHub Actions for CI/CD**, and **OIDC/Keycloak + MFA/passkeys** for **authentication**. Later integrate **real payment gateways**, automated host payouts, maps, notifications, **monitoring with Sentry/OpenTelemetry**, and stronger AI-based recommendations, fraud detection and condition verification.
- **GPS / Location Layer:** PostGIS for spatial queries and geofencing, Mapbox/Google Maps/OpenStreetMap for maps and routing, GPS verification and anti-spoofing, and location intelligence for proximity, demand hotspots, and travel-time matching.
- **AI Layer:** Semantic AI search, personalized recommendations, computer vision for listing/condition verification, AI fraud detection, automated listing generation, RAG-based marketplace assistant, and AI-driven trust and matching.

REFERENCE — Live Demo: <https://spaceloop.onrender.com>